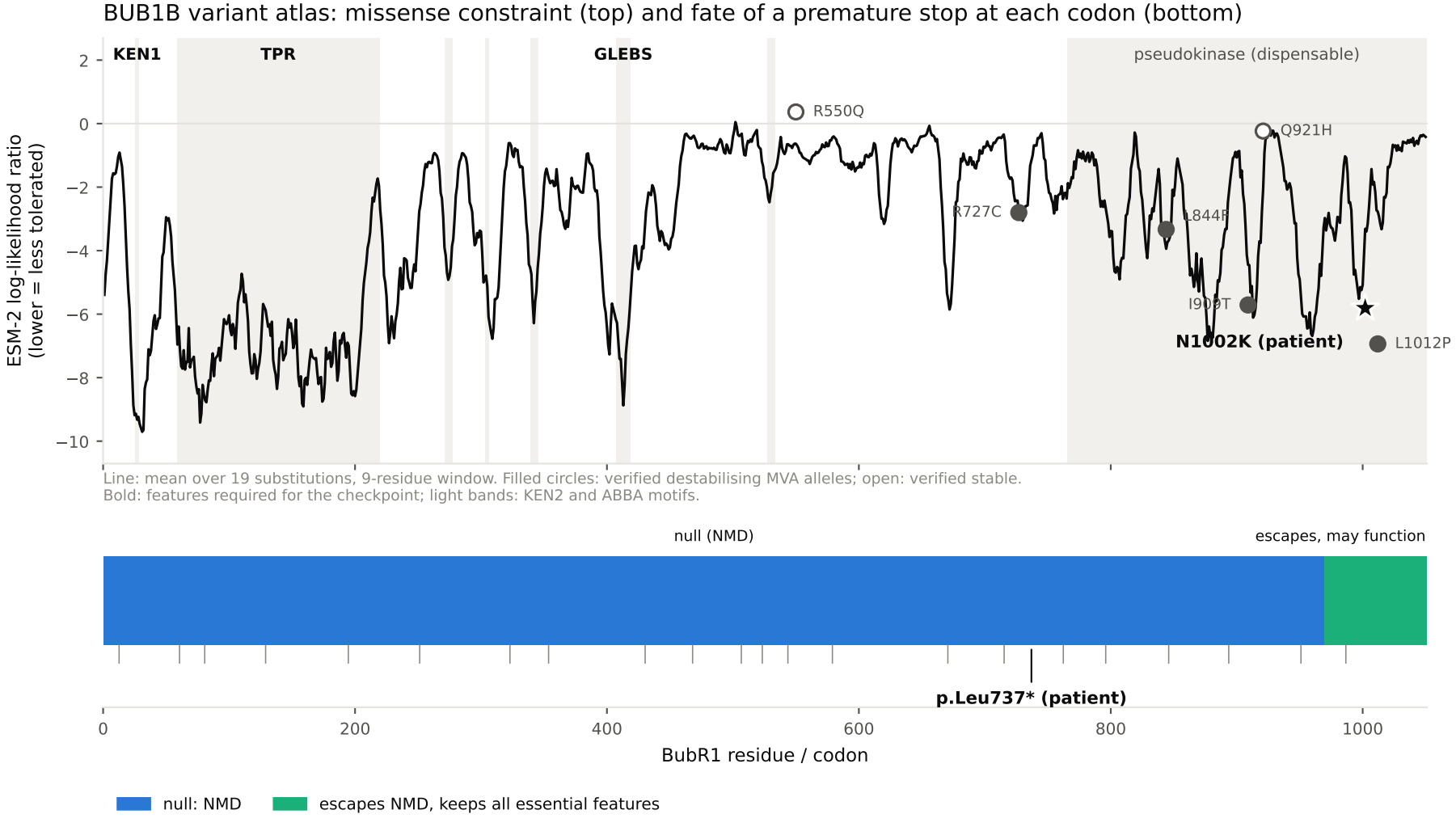
BUB1B variant atlas: missense constraint (top) and fate of a premature stop at each codon (bottom)
KEN1
TPR
GLEBS
pseudokinase (dispensable)
ESM-2 log-likelihood ratio
(lower = less tolerated)
2
0
−2
−4
−6
−8
−10
R550Q
Q921H
R727C
L844F
I909T
N1002K (patient)
L1012P
Line: mean over 19 substitutions, 9-residue window. Filled circles: verified destabilising MVA alleles; open: verified stable.
Bold: features required for the checkpoint; light bands: KEN2 and ABBA motifs.
null (NMD)
escapes, may function
p.Leu737* (patient)
0
200
400
600
800
1000
BubR1 residue / codon
null: NMD
escapes NMD, keeps all essential features